Example 2.1: The following data represent the number of e.mails received by an organisation in different days of a year.

15 18 22 27 30 35 40 18 22 24, 26, 45 30 17 12 32 27 41 25
15 10 12 27 26 21 20 30 28 22 25 28 42 44 48 37 25 23 24 19
20 17 13 11 15 17 32 12 14 18 11

(i) Represent the above observations by a frequency table.

(ii) Find number of days in which e.mails received are 25 and above.

(iii) Find percentage of days in which e.mails received are less than 30.

Solution: (i) Here highest value and lowest value of observations are 48 and 10, respectively. There are N = 50 observations.

$R = x_N - x_1 = 48 - 10 = 38$, where

x_N = highest value of observations;

x_i = Lowest value of observations;

k = No of classes

$= 1 + 3.3 \log N = 1 + 3.3\log 50 = 6.6 \approx 7$

h = width of the class $= \frac{R}{k} = \frac{38}{7} = 5.4 \approx 5$

Title : Frequency table of the number of e. mails received in different days.

Class interval of e. mails	Tally Marks	Frequency f	Cumulative Frequency, c. f.	
10–15	~~////~~ ///	8	8	50
15–20	~~////~~ ~~////~~	10	18	42
20–25	~~////~~ ////	9	27	32
25– 30	~~////~~ ~~////~~	10	37	23
30–35	~~////~~	5	42	13
35–40	//	2	44	8
40^+	~~////~~ /	6	50	6
Total		N = Σf = 50		

(ii) There are 23 days in which e. mails received are 25 and above [c. f. corresponding to calss 25–30 calculated from bottom of the table]

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(iii) There are 37 days in which e. mails received are less than 30 [c. f calculated from top of the table corresponding to class 25–30]. Percentage of observations less than 30 is

$\frac{37}{50} \times 100 = 74\%$

Example 2.2 The following are the amount of diastolic blood pressure (mmhg$_g$) of level of some persons:

Diastolic blood pressure (x) : 70, 72, 78, 85, 82, 80, 65, 77, 75, 70, 72, 82, 74, 76, 85, 88, 90, 82, 66, 70, 72, 74, 75, 80, 75, 90, 88, 68, 70, 78

(i) Represent the above observations by a frequency table.

(ii) Calculate number of persons who have blood pressure level less than 80.

(iii) Calculate percentage of persons who have blood pressure level 75 and above.

Solution: (i) Let us use the value of k = number of classes arbitrarity so that cumulative frequency provides the answer to question No. (ii) and (iii). Here classes are used in such a way that 75 and 80 are in the lower and upper end. respectively, of classes.

Title: Frequency distribution of the blood pressure levels of different persons.

Class interval of blood pressure	Tally Marks	Frequency f	Cumulative Frequency, c. f.	
65–70	///	3	3	30
70–75	~~////~~ ////	9	12	27
75–80	~~////~~ //	7	19	18
80–85	~~////~~	5	24	11
85–90	////	4	28	6
90^+	//	2	30	2
Total	N = Σf = 30			

(i) There are 19 persons [c. f. corresponding to the class 75-80, where c. f is calculated from top of the frequency table.'] who have blood pressure less than 80.

(ii) There are 11 persons [c.f. corresponding to the class 80–85, where c.f. is calculated from the bottom of the class] who have blood pressure 80 and above. So, $(11/30 \times 100) = 36.7\%$

Example 2.3: The following are the number of nurses working in different hospitals :

Number of nurses (x) : 12, 18, 15, 22, 20, 16, 15, 18, 17, 20, 16, 15, 20, 22, 20.

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Represent the data by a discrete type frequency table.

Solution : Discrete type frequency table is one, where number of observations is shown against each value of the variable.

Title : Frequency distribution of number of nurses working in hospitals.

No. of nurses	Tally Marks	Frequency, f (No. of hospitals	Cumulative Frequency, c. f.	
			From top	From bottom
12	/	1	1	15
15	///	3	4	14
16	//	2	6	11
17	/	1	7	9
18	//	2	9	8
20	////	4	13	6
22	//	2	15	2
Total		$N = \Sigma f = 15$		

Bivariate Frequency Table: The frequency tables shown in Example 2.1, Example 2.2 and in Example 2.3 are for presenting the values of one variable observed from units under investigation. These frequency tables are called uni-variate frequency tables. From each unit more than one variable can also be observed. If two variables are observed from each unit, the number of observations of certain level of one variable corresponding to the level of another variable can also be shown by representing the levels of one variable in column and levels of another variable in row. Such a table is called bi-variate frequency table.

Example 2.4: The following are the gestation period (x in days) and birth-weight (y in kg) of some new born babies of some mothers :

x :	260	265	250	255	260	262	270	268	250	245
y :	3.0	3.2	2.8	2.6	2.8	3.2	3.5	3.0	2.5	2.7
x :	252	255	258	250	255	260	265	270	262	260
y :	3.1	3.4	2.5	2.2	2.6	3.2	3.4	3.2	2.5	2.5
x :	245	248	250	255	265	270	264	258	255	240
y :	2.4	2.5	3.0	2.8	3.0	3.0	2.9	3.0	2.4	2.0

Represent the data by a bi-variate frequency table.

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Solution:

Title: The distribution of number of mothers according to their gestation period and birth-weight of baby.

Class interval of gestation period (x)		Class interval of birth–weight (y)				Frequence f_x
		2.0–2.4	2.4–2.8	2,8–3.2	3.2–3.6	
240–245	Tally Marks Frequency	/ 1				1
245–250	Tally Marks Frequency		/// 3			3
250–255	Tally Marks Frequency	/ 1	/ 1	/// 3		5
255–260	Tally Marks Frequency		//// 4	// 2	/ 1	7
260–265	Tally Marks Frequency		// 2	/// 3	// 2	7
265–270	Tally Marks Frequency			// 2	// 2	4
270–275	Tally Marks Frequency			/ 1	// 2	3
Frequency, f_y		2	10	11	7	30

Example 2.5.: The following data represent the number of married couples by their duration of marriage (x in years) and number of ever born children (y).

x :	18	2	1	5	10	12	14	12	18	20	24	18	20	25	27	25
y :	6	1	0	2	2	2	4	4	3	2	6	3	4	8	6	4
x :	10	12	13	14	10	10	9	8	4	6	5	8	10	5	6	6
y :	2	2	4	4	4	2	2	3	1	2	1	1	4	2	2	3
x :	17	18	20	7	9	10	10	12	14	24	27	18	30	16	28	25
y :	3	3	6	2	2	4	2	4	5	8	10	6	10	6	10	10
x :	4	6	28	30	24	23	20	21	22	22	20	18	20	18	20	10
y :	1	1	9	6	6	6	7	5	4	6	3	2	7	4	5	2
x :	3	2	2	2	4	4	4	5	6	7	7	6	5	5	9	10
y :	1	1	0	1	1	1	2	2	2	2	1	1	1	1	2	2
x :	14	16	11	9	7	10	12	14	16	9	10	6	16	18	12	10
y :	3	4	4	3	2	3	3	4	4	3	3	2	2	4	4	2

Represent the number of married couples in a bivariate frequency table.

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Solution : Title : The distribution of number of married couples according to their duration of marriage and number of ever born children.

Class Interval of duration of marriage		Class interval of number of ever born cildren					Frequency fx
		0—2	2—4	4—6	6—8	8+	
<5	Tally marks	~~||||~~ ~~||||~~	|	—	—	—	
	Frequency	10	1				11
5—10	Tally marks	~~||||~~ ||	~~||||~~ ~~||||~~ ~~||||~~ ||				24
	Frequency	7	17				
10—15	Tally marks	—	~~||||~~ ~~||||~~ |||	~~||||~~ ~~||||~~ ||	—	—	25
	Frequency		13	12			
15—20	Tally marks	—	~~||||~~ |	||||	|||	—	13
	Frequency		6	4	3		
20—25	Tally marks	—	||	||||	~~||||~~ ||	|	14
	Frequency		2	4	7	1	
25+	Tally marks	—	—	|	||	~~||||~~ |	9
	Frequency			1	2	6	
	Frequency, fy	17	39	21	12	7	96

Example 2.7: The following are the number of nurses working in different clinics in a city.

Clinics	C_1	C_2	C_3	C_4	Total
Male nurse	12	8	5	6	31
Female nurse	18	12	20	12	62
Total	30	20	25	18	93

Represent the data by (i) bar diagram, (ii) component bar diagram.

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Solution:

(i) Title: Bar diagram representing the number of nurses working in clinics.

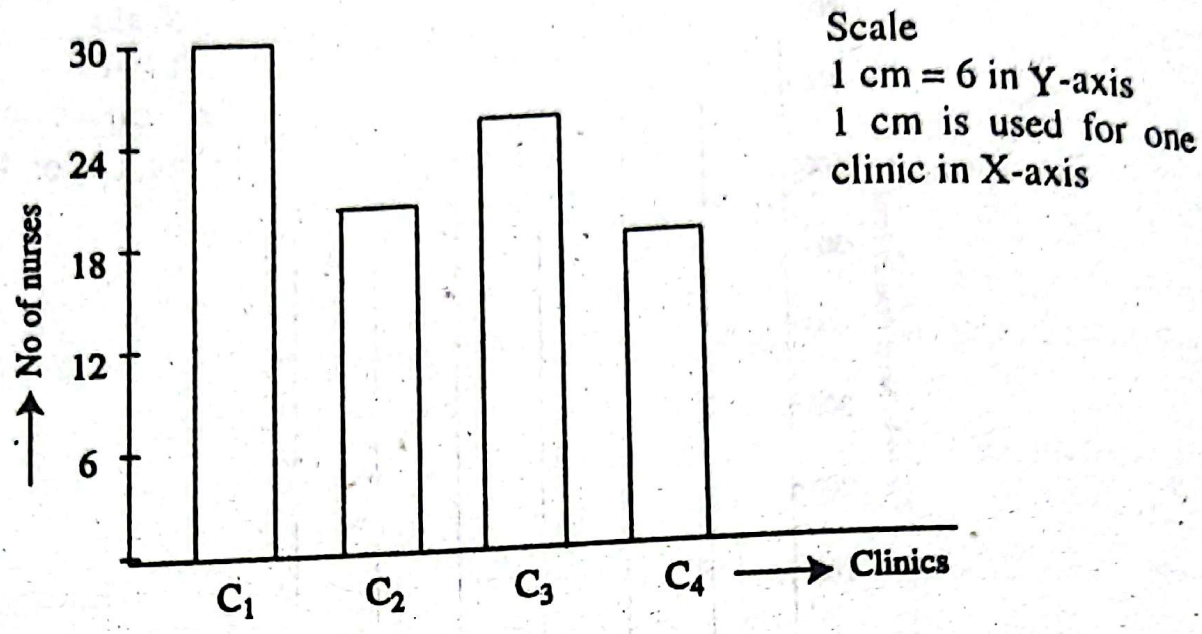

Fig. 2.2 : Bar Diagram

(ii) Title : Component bar diagram to present the male and female nurses working in clinics.

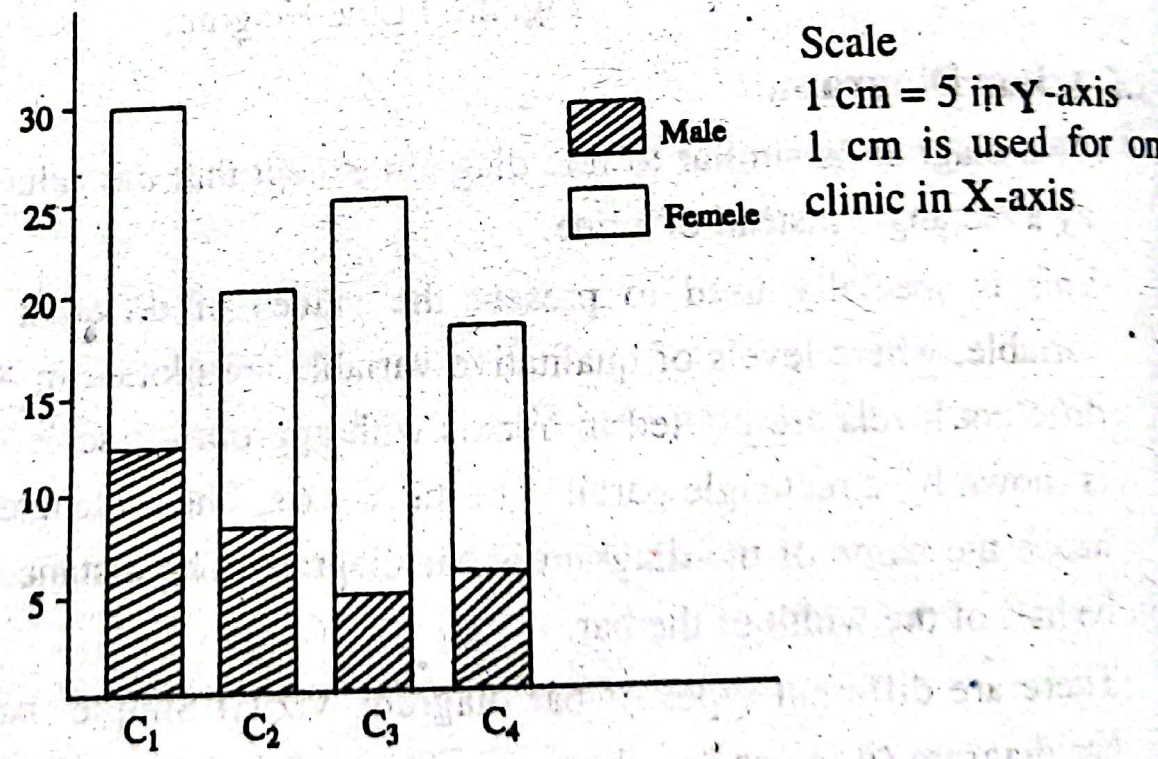

Fig. 2.3 : Component bar diagram

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